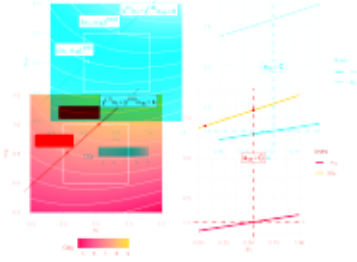
Linear Support Vector Machines

Support Vector Machine Training

Learning goals

- Know that the SVM problem is differentiable
- Know how to optimize the SVM problem in the primal via subgradient descent
- Know how to optimize SVM in the dual formulation via pairwise coordinate ascent



SUPPORT VECTOR MACHINE TRAINING

- Until now, we have ignored the issue of solving the various convex optimization problems.
- The first question is whether we should solve the **primal** or the **dual problem**.
- In the literature SVMs are usually trained in the dual.
- However, SVMs can be trained both in the primal and the dual – each approach has its advantages and disadvantages.
- It is not easy to create an efficient SVM solver, and often specialized appraoches have been developed, we only cover basic ideas here.

TRAINING SVM IN THE PRIMAL

Unconstrained formulation of soft-margin SVM:

$$\min_{\boldsymbol{\theta},\theta_0} \quad \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2 + \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

where $L(y, f) = \max(0, 1 - yf)$ and $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^T\mathbf{x} + \theta_0$.
(We inconsequentially changed the regularization constant.)

We cannot directly use GD, as the above is not differentiable.

Solutions:

1. Use smoothed loss (squared hinge, huber), then do GD.
   NB: Will not create a sparse SVM if we do not add extra tricks.
2. Use **subgradient** methods.
3. Do stochastic subgradient descent.
   Pegasos: Primal Estimated sub-GrAdient SOlver for SVM.

PEGASOS: SSGD IN THE PRIMAL

Approximate the risk by a stochastic 1-sample version:

$$\frac{\lambda}{2}\|\boldsymbol{\theta}\|^2 + L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

With: $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^T\mathbf{x} + \theta_0$ and $L(y, f) = \max(0, 1 - yf)$
The subgradient for $\boldsymbol{\theta}$ is $\lambda\boldsymbol{\theta} - y^{(i)}\mathbf{x}^{(i)}\mathbb{I}_{yf<1}$

Stochastic subgradient descent (without intercept θ_0)

```
for t = 1, 2, ... do
    Pick step size α
    Randomly pick an index i
    If y^(i) f(x^(i)) < 1 set θ^[t+1] = (1 − λα)θ^[t] + α y^(i) x^(i)
    If y^(i) f(x^(i)) ≥ 1 set θ^[t+1] = (1 − λα)θ^[t]
end for
```

Note the weight decay due to the L2-regularization.

TRAINING SVM IN THE DUAL

The dual problem of the soft-margin SVM is

$$\max_\alpha \quad \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y^{(i)} y^{(j)} \left\langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \right\rangle$$

$$\text{s.t.} \quad 0 \leq \alpha_i \leq C \quad \sum_{i=1}^{n} \alpha_i y^{(i)} = 0$$

We could solve this problem using coordinate ascent. That means we optimize w.r.t. α_1, for example, while holding $\alpha_2, \ldots, \alpha_n$ fixed.

But: We cannot make any progress since α_1 is determined by $\sum_{i=1}^{n} \alpha_i y^{(i)} = 0$!

TRAINING SVM IN THE DUAL

Solution: Update two variables simultaneously

$$\max_\alpha \quad \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y^{(i)}y^{(j)}\left\langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right\rangle$$

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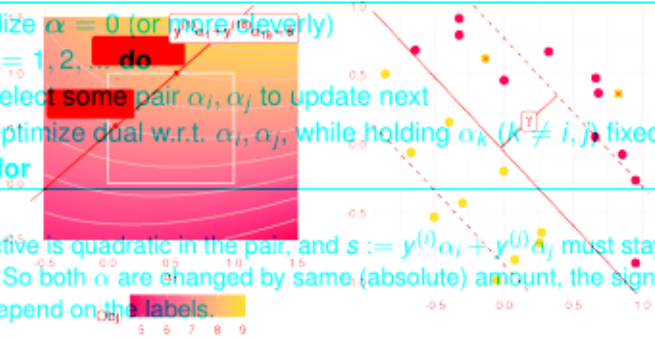
We move on the linear constraint until the pair-optimum or the bounday (here: $C = 1$).

Pairwise coordinate ascent in the dual

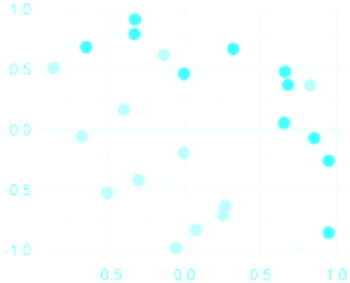
1: Initialize $\alpha = 0$ (or more cleverly)
2: **for** $t = 1, 2, \ldots$ **do**
3: Select some pair α_i, α_j to update next
4: Optimize dual w.r.t. α_i, α_j, while holding α_k $(k \neq i, j)$ fixed
5: **end for**

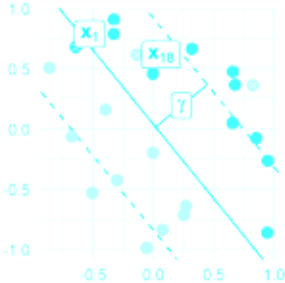


The objective is quadratic in the pair, and $s := y^{(i)}\alpha_i + y^{(j)}\alpha_j$ must stay constant. So both α are changed by same (absolute) amount, the signs of the change depend on the labels.

TRAINING SVM IN THE DUAL

Assume we are in a valid state, $0 \leq \alpha_i \leq C$. Then we chose[1] two observations (encircled in red) for the next iteration. Note they have opposite labels so the sign of their change is equal.





TRAINING SVM IN THE DUAL

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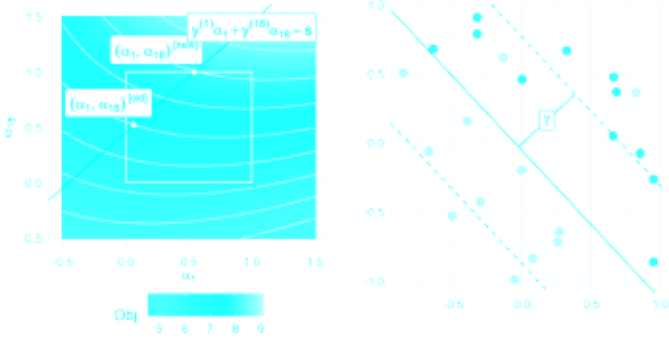
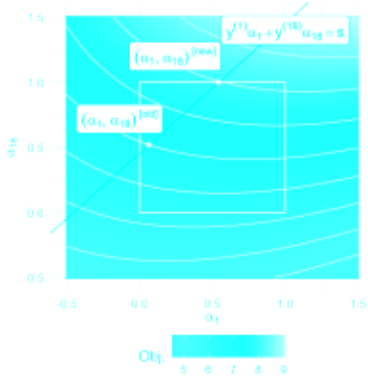
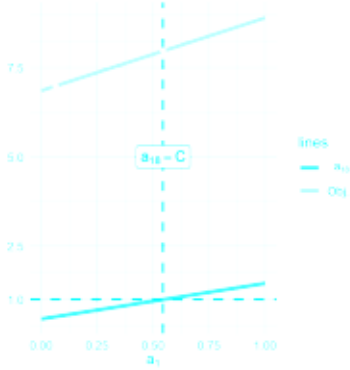


TRAINING SVM IN THE DUAL

Sequential Minimal Optimization (SMO) exploits the fact that effectively we only need to solve a one-dimensional quadratic problem, over in interval, for which an analytical solution exists.